ADITYA CHAVAN

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Computer Science undergraduate specializing in systems programming and backend development. Built a complete in-memory Unix-like virtual file system in C++ featuring custom inode tables, file descriptor management, and seven core system calls. Strong foundation in Linux internals, C/C++, Java/Spring Boot, and clean layered architecture. Focused on correctness, simplicity, and production-ready code.

EDUCATION

Bachelor of Science, Computer Science — 7.8 CGPA 2023 – 2026

Savitribai Phule Pune University, Pune, India

EXPERIENCE

Iron & Logic — Founder & Web Architect Aug 2026 – Present

Pune, India

- Built and deployed websites for local businesses using React, Tailwind CSS, and Vercel — optimized for mobile and fast load times.
- Handled full project ownership — client research, UI/UX, deployment, and direct B2B sales to local business owners.
- Identified digital infrastructure gaps in local markets and pitched custom web solutions directly to business owners.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, JavaScript
Systems: Linux, UNIX internals, Bash/Shell scripting, OS concepts (processes, memory, file systems)
Backend: Spring Boot, REST APIs, Layered Architecture (Controller–Service–Repository)
Frontend: React, Tailwind CSS
Databases: MySQL, PostgreSQL, MongoDB, SQLite, JDBC
Tools: Git, Docker, GitHub Actions, Vercel, Cloudflare, Vim, IntelliJ IDEA, VS Code

TECHNICAL PROJECTS

Customized Virtual File System (CVFS) Feb 2025

C++ / Linux

https://github.com/aditya8242/CustomizedVirtualFileSystem

- Designed and implemented a fully in-memory Unix-like virtual file system from scratch, including custom inode tables, User File Descriptor Table (UFDT), and precise file offset tracking.
- Built seven core system calls (creat, open, read, write, unlink, stat, ls) modeled on UNIX UFS internals, with permission handling, structured error codes, and a clean CLI interface.

Concurrent FTP Server Feb 2026

C / Sockets / Linux

https://github.com/aditya8242/ConcurrentFTPServer

- Built a TCP file-transfer server using fork-per-connection architecture, allowing multiple clients to download files in parallel via isolated child processes.
- Designed a custom line-based protocol (filename request → size header → raw byte stream) for reliable file transfer over raw sockets.

IronVault — File Archiver with Encryption Jul 2026

Java / Swing

https://github.com/aditya8242/IronVault

- Built a file archiver that packs an entire directory into a single binary archive using a custom header format and repeating XOR encryption keyed to a user-supplied password.
- Supports 18 file formats including documents, images, source code, and media. Ships with a Swing GUI — no external dependencies.

EduTrack — Classroom & Student Management Portal Jan 2025

Java / Spring Boot / MongoDB / Docker

https://github.com/aditya8242/EduTrack · Live: https://edutrack-3426.onrender.com

- Developed a production-ready Spring Boot backend with RESTful CRUD APIs for courses and fee management, using MongoDB and clean Controller–Service–Repository architecture.
- Dockerized and deployed on Render with MongoDB Atlas — live at edutrack-3426.onrender.com

OPEN SOURCE

- Contributed Doxygen-compatible documentation to ccgargantua/ini (C library). Iterated through maintainer review, verified IDE rendering, and confirmed clean Doxygen build output. PR #24 merged.

CERTIFICATIONS

PostgreSQL Developer — Infosys Springboard (2025) · Basics of Python — Infosys Springboard (2025)